EPJ Photovoltaics **15**, 41 (2024)
© I. Al-Masalha et al., Published by EDP Sciences, 2024
https://doi.org/10.1051/epjpv/2024037

Available online at:
www.epj-pv.org

ORIGINAL ARTICLE OPEN ACCESS

Improving photovoltaic module efficiency using water sprinklers, air fans, and combined cooling systems

Ismail Al-Masalha, Abdel Salam Alsabagh*, Omar Badran, Naim Alkawaldeh, Taiseer M. Abu-Rahmeh, and Aiman Al Alawin

Faculty of Engineering Technology, Al-Balqa Applied University, Amman, Jordan

Received: 16 March 2024 / Accepted: 21 October 2024

Abstract. This research investigates the essential role of cooling systems in optimizing the performance of photovoltaic panels, particularly in hot climates. Elevated temperatures on the back surface of photovoltaic panels pose a challenge, potentially reducing electrical output and overall efficiency. To address this, a cooling system employing water spray and air was proposed and examined across three scenarios. Results show a consistent reduction in panel temperature with the implemented cooling systems. The drop in temperature of 24 °C (40%) and increase in the output power of 13% indicates its effectiveness, with peak efficiency observed during high ambient temperatures. When water cooling was used, the percentage drop in temperature ranged from approximately 20.5% to 35.4%. The air-cooling system achieves a percentage drop in temperature, ranging from approximately 6 to 26%. Hybrid cooling proves most effective in substantially reducing panel temperature compared to water or air alone. A detailed examination of output power reveals consistent and significant increases with cooling throughout the day. A significant higher percentage power increase is achieved during peak solar radiation hours highlighting the effectiveness of the system during intense sunlight exposure. The cumulative effect reinforces the practical significance of the cooling system, showing its potential to enhance daily energy yield. Efficiency trends consistently indicate higher efficiency with cooling compared to without cooling. The hybrid system, which appears as a promising solution for diverse environmental conditions cases, improves the overall efficiency by up to 2%.

Keywords: Solar energy / photovoltaic / water cooling / fans cooling / water sprays

1 Introduction

Solar panels are innovative devices that convert sunlight into usable electricity, that have gained widespread popularity as a renewable energy source. Technological advancements and concerns about the environmental impact of fossil fuels have propelled their adoption.

Beyond mitigating carbon emissions, solar panels offer long-term cost savings for homeowners and are versatile enough to be installed on rooftops, cars, and even satellites. They represent a significant investment for homeowners interested in reducing their carbon footprint and lowering monthly utility bills. With minimal maintenance requirements and a lifespan of decades, solar panels emerge as a sustainable choice for powering homes. Furthermore, Jordan and many countries provide tax incentives, making the transition to solar energy more affordable. Thus, solar energy stands out as one of the most accessible and sustainable sources of electricity. Photovoltaic (PV) systems, utilizing sunlight to generate electricity, contribute to renewable and clean energy solutions, thereby curbing global warming.

Despite their numerous advantages, solar panels encounter challenges, particularly during the summer when an increase in temperature can lead to reduced performance. To address this issue, various cooling systems have been developed to lower panel temperatures, enhancing efficiency and productivity.

Al-Jamea et al. [1] have conducted experimental work to improve the performance of PV panels by adopting two types of water-cooling systems, namely immersion and spraying. A reduction in the temperature by 60% was noted with a 59% increase in power output and an 8% increase in efficiency. Siecker et al. [2] have conducted a review paper discussing different methods that may be used to mitigate the negative sequences due to excessive increases in the PV temperature while attempting to enhance efficiency. Their paper addressed different cooling techniques like Floating Tracking Concentrating Cooling systems (FTCC); using water spraying for cooling hybrid solar Photovoltaic/Thermal (PV/T) systems; PV cooling by immersing

* e-mail: alsabagh@bau.edu.jo

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

techniques; and the use of forced circulated water and air for PV cooling purposes. A comprehensive review of modern studies concerning the design and efficient operation of Photovoltaic/Thermal systems was conducted by Abdelrazik et al. [3]. The authors emphasized the use of nanofluids and phase-change materials to enhance PV cooling and storage. They focused on the application of cooling fluid on both sides of the solar panel, along with optical filtration using nanofluids, which resulted in impressive improvements. Bahaidarah et al. [4] have addressed the importance of uniform cooling of the solar panels by sightseeing the effects of the non-uniformity. The comparison between the uniform and non-uniform cooling methods was discussed by conducting an experimental case study, the experiment concluded that using immersion cooling is an optimal solution for solar panels and can remarkably reduce the temperature to 25–45 °C for the concentrated Photovoltaic (CPV) systems. The best case of temperature non-uniformity of 3 °C was achieved by the use of heat pipes by which the cell temperature was reduced to 32 °C while using passive cooling by heat sinks reduced the cell temperature to 37 °C. According to their works, the most effective cooling method was active cooling by using impingement cooling and hybrid microchannel-impingement cooling by which the temperature decreased to 30 °C. A water-air hybrid cooling system was proposed by Azmi et al. [5], and the results have proven an increase in the overall efficiency by 4.5%.

The various PV passive cooling techniques were explained and discussed by Mahdavi et al. [6] taking into consideration geographical and environmental conditions, the air ventilation method was found to be the most economical and practical way. The authors have highly recommended floating PV for the case of water cooling. Panda et al. [7] and Zubeer et al. [8] have presented and explained the most advanced PV cooling techniques including detailed discussions of their characteristics and capacities. An in-depth study of different PV cooling methods was addressed by Dwivedi et al. [9]. Their paper discussed various techniques for excessive heating problems. Furthermore, improvement and optimization of different types of cooling systems were provided. Nižetić et al. [10] have performed an experimental setup of a PV water spray-cooling system on both sides for a typical Mediterranean climate. Their results showed an increase in the output efficiency and electric power output by 16.3% and 14.1% respectively. A drastic reduction in temperature from 54 °C to 24 °C was reported when both sides of the PV panel were cooled. Moharram et al. [11] have employed a mathematical model to predict the best timing regime for water spray cooling provided to typical PV panels in a desert area in Egypt. Their model was intended to reduce the amount of water used for cooling purposes. The model was experimentally validated to cool down the PV panels to their normal operating temperature at 35 °C, the highest output energy found to be when cooling starts at the maximum allowable temperature of 45 °C.

However, problems with photovoltaic systems include hail, dust, and their surface temperature, which ultimately reduces the efficiency of solar energy conversion. Environmental issues influencing the surface temperature of a PV module involve various weather conditions, wind velocity, relative humidity, gathered dust, and solar radiation intensity as was pointed out by Jathar et al. [12]. Increasing the surface temperature of the solar module by just 1 °C can result in a 0.5% decrease in yield. The excess solar energy not converted into electricity is transformed into heat, affecting overall conversion efficiency. To preserve solar energy resources, intensive research has focused on solar cooling technologies. Various cooling systems, including liquid cooling and active air cooling, were used to ensure solar panels' optimal and efficient functioning, establishing solar energy as a reliable power source. Implementing these cooling systems enhances the effectiveness and lifespan of solar cells, making solar energy an increasingly viable long-term alternative.

Efficient cooling of solar panels is critical to maintaining peak performance and longevity. Excessive heat can diminish the efficiency of solar cells, leading to decreased energy production and potential damage to the entire system. Water cooling systems emerge as an effective means to regulate the temperature of solar panels, and maintain efficiency. These systems circulate water through tubes underneath or above the panels, absorbing heat and carrying it away from the cells. One notable advantage is that water cooling can be seamlessly integrated into hybrid cooling systems, combining air and water to maximize energy output. Overall, water cooling proves to be a reliable method for managing temperatures in solar panel installations, ensuring maximum efficiency and power output. The combination of air and water for cooling solar cells, known as a hybrid cooling system, is a common technique to enhance the efficiency and longevity of photovoltaic (PV) systems. This approach leverages the advantages of both air and water-cooling methods, Koohestani et al. [13].

Natural convection, a passive cooling strategy, utilizes the principle that hot air rises and cooler air sinks, creating a natural circulation of air around the panels. Natural convection cooling capitalizes on the temperature difference between the solar panel surface and the surrounding air. As solar panels absorb sunlight and heat up, the air near the panel's surface becomes warmer and less dense. This warm air rises, creating a flow that carries heat away from the panel. Cooler air from the surroundings then replaces the rising warm air, establishing a continuous cycle of air movement, but this process is weak and can't provide proper cooling to the PV, while forced convection cooling, an active method, involves mechanically circulating air over PV solar panels, enhancing heat dissipation and maintaining lower temperatures. This active airflow significantly improves efficiency and prolongs the lifespan of PV panels. Forced convection cooling for PV solar panels is a sophisticated method designed to actively regulate and control the temperature. It employs mechanical components, such as fans or blowers, strategically positioned to facilitate the continuous circulation of ambient air over the PV panels. Researchers have applied various approaches to cool PV modules, avoiding excessive heating and reducing panel temperature, resulting in increased power production, as explained in the following studies.

Shalaby et al. [14] have simultaneously examined two identical PV models with and without cooling, an improvement in the generated power by 14.1% and an increase by 2.4% for the electrical efficiency were noted by implementing preheating the feed water of reverse osmosis desalination cooling system. Kazem et al. [15] and Arifin et al. [16] investigated the impact of high solar irradiation on PV panels and proposed an air-cooled heat sink design to reduce their operating temperature. The effectiveness of the heat sink was analyzed using both numerical and experimental methods, and it was found to substantially decrease the operating temperature of the PV panel while improving its electrical performance. The average temperature of the PV panel was reduced from 85.3 °C to 72.8 °C by the heat sink, resulting in a 10% increase in open-circuit voltage and an 18.67% increase in maximum power. According to Palumbo [17], the implementation of forced convection through fins on the surface of a PV panel reduced its average temperature and resulted in a 10% increase in power output. Cuce et al. [18] investigated the performance of polycrystalline PV cells under various circumstances. Two PV cells were used: one with an aluminum heat sink and thermal paste, and the other without a heat sink. The cells were exposed to illumination ranging from 200 to 800 W/m^2. The use of passive cooling through a heat sink resulted in a 9% improvement in electricity efficiency. Mazón-Hernández et al. [19] observed that the depth of the flow channel underneath PV cells has a substantial influence on passive cooling, particularly for larger PV surfaces (1.95 m^2). They found that for a length-to-depth ratio of 0.085, the heat generated by the PV module increased by 5–6 °C compared to a PV module on a uniform mount. Variations in temperatures were observed to increase with increasing insolation. Conversely, passive flow channels may have the opposite effect on PV cooling. Nader et al. [20] revealed that at a temperature of 42 °C, solar panel efficiency varied from 10% to 15% under typical circumstances. However, after implementing a cooling system, the temperature of the solar cells decreased to 20 °C, resulting in a 7% increase in efficiency. Furthermore, the output power increased by 30%, with a peak efficiency of 32.5% at midday. According to Rakino et al. [21], the proposed cooling system outperformed the comparative systems, which merely featured only a heat sink or water cooling. The proposed system resulted in a much lower average surface temperature, which was 12.66% lower than that of the basic solar panel. This resulted in higher output voltage, load current, and output power of 21.49%, 4.66%, and 47.71% respectively compared to the plain solar panel. The proposed system produced 8.34% more output power than the compared panel. Popovici et al. [22] examined the reduction in the temperature of PV panels on a clear summer day using various configurations of wall heat sinks with air and passive cooling techniques. The numerical analysis revealed that the highest temperature of the panel at an angle of 45° was lower than that at an angle of 135°. Additionally, the employment of heat fins resulted in a 6.97% and 7.55% increase in the highest power produced by the PV panel as compared to the reference case for rib angles of 90° and 45°, respectively. Farhana et al. [23] investigated the effect of operating temperature variation on the electrical performance of a PV module with and without an active cooling system. The experiment includes employing two crystal PV panels with 13% maximum efficiency working under standard conditions (1000 W/m^2, 25 °C), one of the panels was used as a reference while the other was fixed at the base as a heat sink. To heat the fins, a DC brushless fan was connected. As a result of the cooling system, the PV temperature exceeded the ambient temperature by 30% with a 70% increase in the temperature in the absence of the system. furthermore, the open circuit voltage (V_{oc}) of the PV panel was somewhat higher than that of the PV panel without such a cooling system. Sandhu et al. [24] examined two types of PV modules, namely glass-to-glass and glass-to-tedlar, with and without a duct beneath the panel in New Delhi, India. The dimensions of the duct were 0.605 m × 1.0 m × 0.04 m, and the PV tilt angle was set at 30°. The results indicated that the glass-to-glass modules with a duct yielded a higher electrical efficiency and outlet air temperature among all the cases. Furthermore, the percentage difference in the electrical efficiency of glass-to-glass type PV modules with and without a duct was 0.66%, which can be considered significant when employed in large PV plants. Additionally, the annual average efficiency of glass-to-glass type PV modules with a duct was 10.41%, compared to 9.75% without a duct. Salmanzadeh et al. [25] studied the integrated photovoltaic panels coupled with air thermal collectors to enhance the energy efficiency of a mechanical ventilation system in Kerman City, Iran. The system was designed to cool the PV units during the summer and use the preheated air underneath the warm panels for space heating in the winter. The authors claimed that an annual increase in electricity production by 7.2% was achieved. Moreover, an amount of 3400 kWh was recovered for heating space of 10 m^2 of PV panels along with an additional 56 kWh of electricity.

An experimental study was conducted by Dwivedi et al. [26] to analyze the thermal and electrical performance of panels passively cooled along with moist coconut fibers under Malaysian tropical climate conditions. The experimental work was compared with a PV/T system. The surface temperature was reduced by more than 20% for a PV system, while the reduction for the PV/T system was less than 17%. Energy efficiency was increased by 11% and 9% for PV and PV/T systems, respectively. Durez et al. [27] have performed a detailed mathematical model using three different phase change materials (PCMs), namely, RT35, RT21, and RT44 in order to adapt various weather conditions in different parts of the world in the summer. MATLAB was employed in an optimization model suitable for (PCMs), the authors concluded that integrating (PCMs) resulted in a reduction in the surface temperature of the PV panels leading to an increase in the efficiency and electrical output by 6% and 16% respectively. Dwivedi et al. [28] have addressed the various cooling techniques throughout a comprehensive review paper including air-based cooling systems by which the PV temperature could be maintained below 40 °C, the authors also presented other methods like liquid cooling which covers forced water circulation, liquid immersion cooling, water spraying and phase change models (PCMs).

In addition, other techniques were clarified like heat pipe, heat sink, micro channels heat exchanger, Nano-fluid based cooling, and hybrid/ multi-concept cooling systems.

The present research concentrated on the effect of cooling on the performance of PV using simple systems away from any complex interactions. There are many methods were used as explained in the introduction, but many of them seem so costly or difficult to implement in a large application due to their complexity. The cooling processes used in this research were simple in design and erection and they require less maintenance and operation. Despite the significant potential of solar energy as a source of clean and renewable power, the primary issue with photovoltaic cell operation is the loss of electrical energy due to increased cell temperatures, which leads to reduced energy output and electrical efficiency [29–31]. Elevated temperatures raise the resistance within solar cells, diminishing their ability to convert sunlight into electricity efficiently. This effect, known as the temperature coefficient, results in decreased energy production as temperatures rise [32]. Moreover, high temperatures accelerate the degradation of materials used in PV panels, such as semiconductor materials and packaging layers, leading to a decline in overall performance and lifespan, thereby affecting the reliability and long-term viability of solar panels [33]. Excessive heat can also worsen the "hot spot" phenomenon, where localized areas of the panel experience significantly higher temperatures than others. Hot spots can cause irreparable damage to affected cells, reducing the overall efficiency of the PV system [34], with permanent damage caused by hot spots estimated at 15% [35].

1.1 Addressing energy consumption and water usage in cooling systems

In this study, the direct electrical energy consumption of the pump was not considered, as there are alternative methods to supply the cooling system with water without relying on direct electrical input. One such method involves utilizing existing farm irrigation infrastructure to provide cooling water indirectly to the system. By tapping into irrigation networks before water reaches agricultural fields, the need for direct electrical energy consumption or additional water usage from domestic sources, car wash stations, etc., can be eliminated. This approach offers a sustainable solution that leverages existing resources without imposing additional strain on the electrical grid or water supply.

Furthermore, the spray method for cooling proves to be effective, offering comprehensive coverage of the entire photovoltaic panel based on the number of sprays employed. Even in instances where water may not reach certain areas during spraying, the thermal conduction process ensures that these sections maintain lower temperatures. This ensures uniform cooling across the panel surface, maximizing efficiency and performance.

It's worth noting that the sprinkler system is typically installed on the backside of the photovoltaic panel, shielding it from direct exposure to sunlight. As a result, the amount of water evaporation is minimized, contributing to efficient water usage and reducing overall water consumption. This configuration optimizes the cooling process while minimizing water wastage, aligning with sustainability goals and environmental conservation efforts.

By adopting such strategies, we can enhance the efficiency and sustainability of cooling systems for photovoltaic panels, reducing energy consumption and water usage while maximizing performance and longevity. These findings underscore the importance of considering alternative cooling system design and operation approaches, paving the way for more environmentally friendly and resource-efficient solutions in the renewable energy sector.

1.2 Scope of the study

This research focuses on the critical role of cooling systems in enhancing the performance of photovoltaic (PV) panels, specifically in hot climate conditions where elevated temperatures can significantly reduce the electrical output and overall efficiency of the panels [36]. The study investigates three different cooling methods: water spray, air cooling, and a hybrid system combining both. By analyzing these methods, the study aims to determine the most effective cooling solution for maintaining optimal PV panel performance.

The novelty of this study is found in its detailed comparative analysis of cooling methods, the introduction and evaluation of a hybrid cooling system, precise temperature reduction metrics, and the quantification of performance improvements, all of which offer valuable contributions to optimizing PV panel performance in hot climates. Table 1 summarizes what previous authors have done in this research area.

2 Methodology

2.1 Experimental set-up of solar panel cooling

Water cooling is a highly effective method employed to dissipate heat from photovoltaic (PV) modules, enhancing their overall performance. This process entails circulating water through a system of pipes, bringing it into direct contact with the modules through strategically positioned sprinklers. As the water traverses the pipes and is sprayed onto the back of the PV modules, it efficiently absorbs heat, preventing the accumulation of excess heat that could otherwise compromise module efficiency. The water-cooling process can be delineated into two primary stages: heat absorption and heat dissipation. In the initial stage of heat absorption, water is pumped through the pipes and encounters the photovoltaic modules by sprinklers. Through this interaction, the water absorbs heat from the modules, becoming heated in the process. The heated water is then directed back to the tank, where it undergoes a cooling process, facilitating the dissipation of heat.

The implementation of water cooling has shown a remarkable capacity to significantly enhance the efficiency of photovoltaic modules, resulting in a substantial increase in energy output and an extended lifespan. Research indicates that water-cooled modules can yield up to 15%

Table 1. A summary of previous work done in the area of this research.

Authors (Year)	Main findings
Azmi et al. [5]	A water-air hybrid cooling system was proposed, and the results have proven an increase in the overall efficiency by 4.5%.
Panda et al. [7] and Zubeer et al. [8]	Presented and explained the most advanced PV cooling techniques including detailed discussions of their characteristics and capacities.
Shalaby et al. [14]	14.1% improvement in generated power and 2.4% increase in electrical efficiency using water cooling in a reverse osmosis desalination system.
Kazem et al. [15]	The use of air-cooled heat sinks reduces the operating temperature of PV panels and improves performance.
Arifin et al. [16]	Reduced PV panel temperature from 85.3 °C to 72.8 °C with air-cooled heat sink, increasing power by 18.67%.
Palumbo [17]	10% increase in power output using forced convection cooling with fins on PV panels.
Cuce et al. [18]	9% improvement in electricity efficiency using passive cooling with a heat sink.
Mazón-Hernández et al. [19]	Significant impact of flow channel depth on passive cooling effectiveness, particularly for larger PV surfaces.
Nader et al. [20]	7% increase in efficiency and temperature reduction to 20 °C using a cooling system.
Rakino et al. [21]	12.66% lower average surface temperature, leading to a 47.71% increase in power output with the proposed cooling system.
Popovici et al. [22]	Up to 7.55% increase in maximum power output using passive cooling techniques with wall heat sinks.
Farhana et al. [23]	Slightly higher open circuit voltage and improved performance with active cooling system.
Sandhu et al. [24]	Higher electrical efficiency in glass-to-glass PV modules with ducts compared to those without ducts.
Salmanzadeh et al. [25]	7.2% annual increase in electricity production using a ventilation system for cooling PV panels.

more electricity compared to air-cooled counterparts, positioning them as a promising solution for solar power generation [37]. Beyond performance improvements, water cooling also mitigates the risk of module failure due to overheating, a critical consideration in hot climates or during periods of intense solar irradiance when excess heat can cause irreversible damage. By maintaining the temperature of the modules within a safe range, water cooling contributes to the extension of their lifespan and the reduction of maintenance costs.

2.2 Instrumentation

In our research, various equipment was employed for PV cooling using pipes and sprinklers, including water pumps, tanks, and two solar panels. The experiments were conducted in the Amman Region from 9:00 a.m. to 5:00 p.m., with readings recorded every hour. Voltage and current values were measured using a multimeter, and theoretical calculations were performed for power output and efficiency. Each piece of equipment is explained as follows.

2.2.1 The Pump

The 220V AC pump with a power rating of 65 W, as depicted in Figure 1, was employed in the experimental setup. The pump facilitated the movement of water from the tank to the distribution pipe, ensuring an even flow through the holes of the sprinklers onto the photovoltaic panel. Subsequently, the water was directed back into the water tank, completing a cycle for subsequent reuse.

2.2.2 Multimeter

The multimeter, a crucial instrument in our experimental setup, played a pivotal role in measuring electrical parameters. This device was employed to assess voltage and current values during the experiments, providing essential data for the analysis of the photovoltaic panel's performance as shown in Figure 2.

2.2.3 Thermometer

The thermometer was an essential tool utilized in our experimental configuration. It served the purpose of monitoring and recording the temperature variations



Fig. 1. AC pump used in the experiment.



Fig. 2. Multimeter used in the experiment.

during the experiments, contributing valuable data for the comprehensive analysis of the photovoltaic panel system's cooling efficiency and thermal management as shown in Figure 3.

2.2.4 Measurement of temperature reduction

Temperature reduction on the PV panel surface was measured using thermocouples strategically positioned at multiple locations across the panel. These thermocouples recorded the surface temperature continuously throughout the experiment. Figure 4 shows a solar intensity power meter used in the experiment.

Equation (1) represents the maximum power that can be extracted from the PV where I_{mp} and V_{mp} are the maximum current and maximum voltage respectively at maximum power conditions P_{max}, the collector efficiency is given in equation (2) where A_c is the collector area and G is the solar radiation. The maximum decrease in the PV temperature is given in equation (3) whereas equation (4) is used to calculate the maximum increase in the power.

$$P_{max} = I_{mp} V_{mp} \tag{1}$$



Fig. 3. Temperature and humidity digital reader used in the experiment for data collecting.



Fig. 4. Solar intensity power meter.

$$\eta_c = \frac{I_{\max} V_{\max}}{A_c G} \tag{2}$$

$$\%\mathrm{T}_{\mathrm{max,\ decrease}} = \frac{\mathrm{T}_{\mathrm{withoutcooling}} - \mathrm{T}_{\mathrm{cooling}}}{\mathrm{T}_{\mathrm{withoutcooling}}} \times 100\% \tag{3}$$

$$\%\mathrm{P}_{\mathrm{max,\ increase}} = \frac{\mathrm{P}_{\mathrm{cooling}} - \mathrm{P}_{\mathrm{withoutcooling}}}{\mathrm{P}_{\mathrm{withoutcooling}}} \times 100\%. \tag{4}$$

Solar radiation was measured using a solar intensity meter placed adjacent to the solar panel. The pyranometer measures the amount of solar radiation incident on the Earth's surface and provides accurate readings of received solar energy. This measurement was used to calculate the solar irradiance (input) to the solar system, which is an important factor in calculating the electrical efficiency of the solar cells.

The pump was used for what is available in the local market, but it did not work at its full capacity, as it can give approximately 20 L/min, but it was operated at low flow quantities that did not exceed 2 L/min.

2.2.5 Solar panel

Solar panels, commonly referred to as photovoltaic (PV) panels, represent devices meticulously engineered to transform sunlight into electricity, constituting a pivotal element in the utilization of renewable solar energy.

Table 2. The specifications of the solar module.

P_{max}	50 Wp
Vmp	18 V
Imp	2.78 A
Voc	21.6 V
Isc	3.05 A



Fig. 5. Overall test rig of the experimental work.

Solar cells: The fundamental unit of a solar panel is the solar cell. Comprising semiconductor materials, predominantly silicon, these cells undergo specific treatments to establish an electric field. When illuminated by sunlight, the cells undergo excitation, leading to the generation of direct current (DC) electricity. The specifications of the solar module pertinent to our experimental study are delineated in Table 2.

Figure 5 illustrates the configuration of water sprinklers and air fans. In this experimental setup, the water sprinklers and air fans were strategically positioned on the back of the PV panel to implement an effective cooling system. The water sprinklers were designed to evenly distribute water across the back surface of the panel, promoting efficient heat absorption. Simultaneously, the air fans enhance the cooling process by facilitating air circulation around the PV panel. This combined approach aims to mitigate excessive heat buildup, ensuring optimal operating conditions for the solar cells and, consequently, improving overall energy output and efficiency.

Table 3 indicates that each instrument has a distinct range of accuracy. To evaluate uncertainty measurement, an uncertainty analysis was performed on the outputs of the experimental equipment and the corresponding physical quantities. There are some uncertainties associated with power, efficiency, and natural area; thus, the standard calibration method was employed to assess these uncertainties. The uncertainty levels in power, area, and efficiency can be calculated using equations (5) through (7) [38]. Uncertainty values are given in Table 3.

$$\frac{UP}{P} = \sqrt{\left(\frac{UI}{I}\right)^2 + \left(\frac{UV}{V}\right)^2} = 0.004\% \tag{5}$$

$$\frac{UA}{A} = \sqrt{\left(\frac{UW}{W}\right)^2 + \left(\frac{UL}{L}\right)^2} = 0.17\% \tag{6}$$

$$\frac{U\eta}{\eta} = \sqrt{\left(\frac{UP}{P}\right)^2 + \left(\frac{UA}{A}\right)^2 + \left(\frac{UG}{Grad}\right)^2} = 0.2\%. \tag{7}$$

2.3 Performance indicators

The utilization of solar energy for electricity generation through photovoltaic panels is imperative. However, the energy output of these panels is adversely affected by elevated temperatures on the back surface, resulting in a potential reduction in electrical output and overall efficiency. This issue is particularly prominent in hot regions like southern Jordan and Jordan Valley during the summer season, where temperatures are notably high. Therefore, a cooling system is essential to mitigate these temperature-related challenges, especially on the back surfaces of photovoltaic panels. The proposed cooling system involves the application of a water and air spray to effectively lower temperatures. Numerous experiments have been conducted on photovoltaic panels to optimize their performance for efficient electricity generation.

The current research is structured around three distinct scenarios: the first scenario involves the use of water for cooling, the second scenario employs air for cooling, and the third one integrates both water and air through a spraying mechanism and fixed fans applied to the back surface of the photovoltaic panel.

2.3.1 Temperature reduction of PV panels with cooling

The results indicated temperature reduction in the photovoltaic (PV) panel with the implemented cooling system compared to the panel without cooling. The percentage drop in temperature serves as a key metric to assess the effectiveness of the cooling system, and the observations across different hours provide valuable insights. Here are some interpretations of the results: Across all recorded hours, the panel temperature with cooling is consistently lower than the panel temperature without cooling. This consistency suggests that the cooling system effectively mitigates the rise in temperature associated with solar radiation. From 11:00 a.m. to 1:00 p.m., when the ambient temperature is relatively high, the cooling system demonstrates its effectiveness by allowing a higher percentage drop in panel temperature. This implies that the cooling system is particularly beneficial during peak sunlight exposure. Based on Figure 6 the percentage drop in temperature ranges from approximately 20.45% to 35.44% when water cooling was used, emphasizing a substantial reduction in panel temperature when the cooling system is active. This reduction is crucial for maintaining the efficiency and longevity of the photovoltaic panels. Based on Figure 7, the implemented air-cooling system reduces the temperature of the photovoltaic (PV) panel compared to the non-cooled panel. The cooling system achieves a percentage drop in

Table 3. Uncertainties for various used instruments.

Instrument	Measured parameters	Accuracy	Range	Uncertainty
Solar power meter	Intensity of solar radiation	$\pm 1\,W/m^2$	0-2000 W/m^2	$0.57\,W/m^2$
Ammeter	Current	± 1 m A	0-10 A	0.5×10^{-3} A
Voltmeter	Voltage	± 1 mV	0-200 V	0.5×10^{-3} V
Thermocouple	Temperature	± 0.1 °C	–210-200 °C	0.57×10^{-3} °C

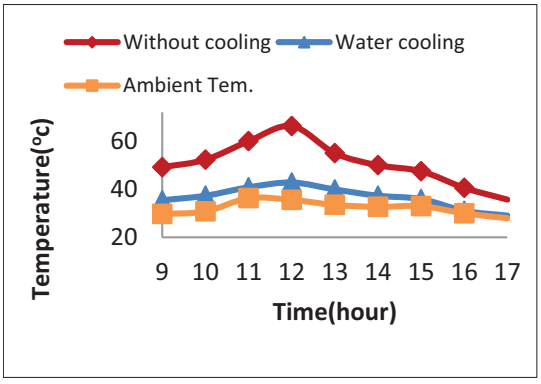


Fig. 6. Comparison of PV temperature when using water sprinklers over time.

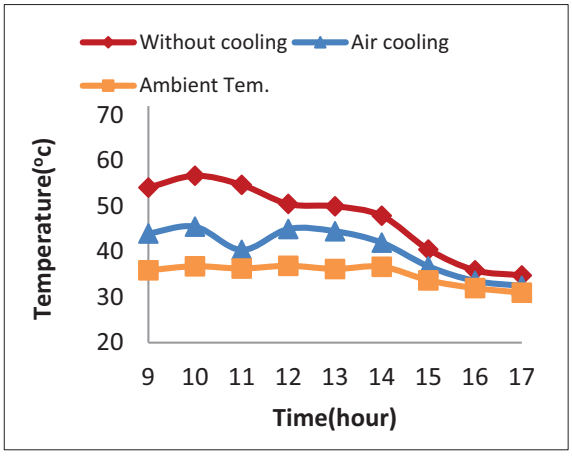


Fig. 7. Comparison of PV temperature when using air cooling over time.

temperature, ranging from approximately 6% to 26%. According to Figure 8, the air-cooling system effectively lowers the temperature of the photovoltaic (PV) panel, with a percentage drop ranging from approximately 30% to 40%.

In general, employing a hybrid cooling system results in a more substantial reduction in the temperature of photovoltaic panels compared to using water alone or air alone.

2.3.2 Power improvement of PV panels with cooling

The comparison between output power without cooling and output power with cooling provides valuable insights into the effectiveness of the cooling system in enhancing the performance of photovoltaic (PV) panels. Here's a detailed explanation and commentary on the results:

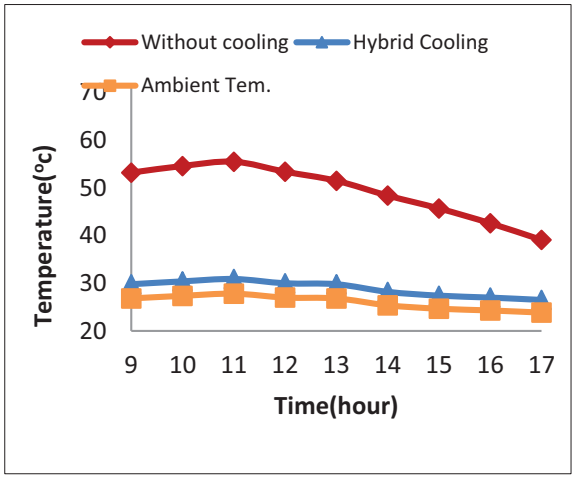


Fig. 8. Comparison of PV temperature when using hybrid cooling over time.

The data shows consistent and significant increases in output power with cooling throughout the day. This indicates that the cooling system is consistently effective in improving power generation across various time intervals. The cooling system's ability to deliver a sustained positive impact on power output is a promising sign of its reliability and efficiency.

During peak solar radiation hours (e.g., 11:00 to 13:00), the percentage increase in output power is notably higher. This aligns with the expectation that the cooling system has a more pronounced effect when the panels are exposed to intense sunlight. The observed peak in power increase during high solar radiation emphasizes the cooling system's efficiency in optimizing power output under challenging conditions.

The cumulative effect of cooling on power production is evident when comparing the total output power without cooling to that with cooling. The consistent increase throughout the day contributes to a substantial overall improvement. The cumulative impact reinforces the practical significance of the cooling system, showcasing its potential to enhance the daily energy yield of the PV panels and showing proper heat transfer from the PV back surface to the surroundings.

The observed variations in output power enhancement highlight the efficiency of different cooling methods. Based on Figure 9, the water cooling exhibits a notable impact, with increases ranging from 3.25% to 12.29%, showing its effectiveness in optimizing power generation. Similarly, the air-cooling system, as illustrated in Figure 10, achieves significant output power improvements, ranging from

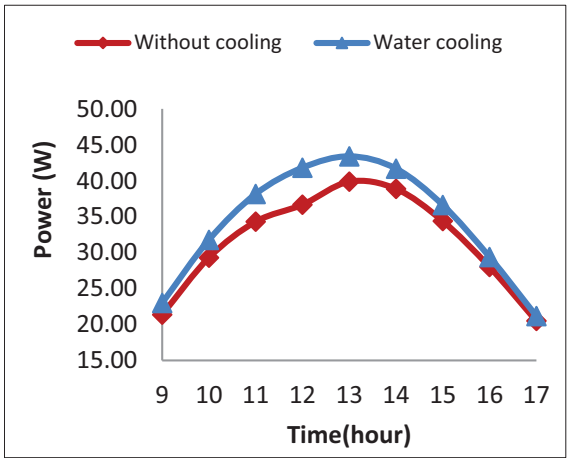


Fig. 9. Comparison of output power when using water sprinklers over time.

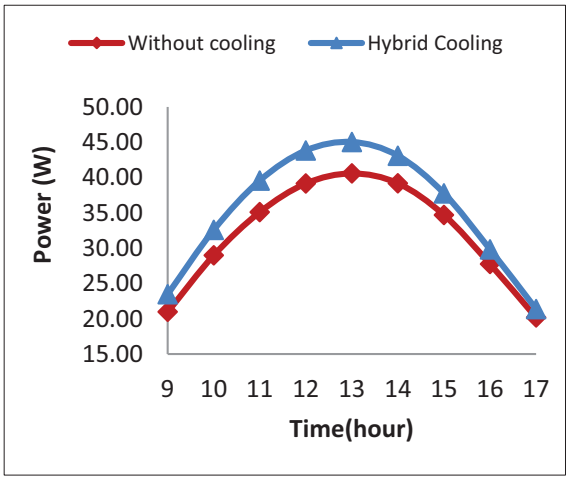


Fig. 11. Comparison of output power when using hybrid cooling over time.

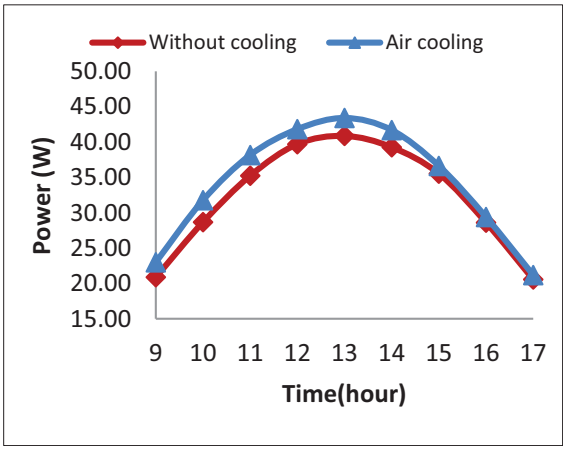


Fig. 10. Comparison of output power when using air cooling over time.

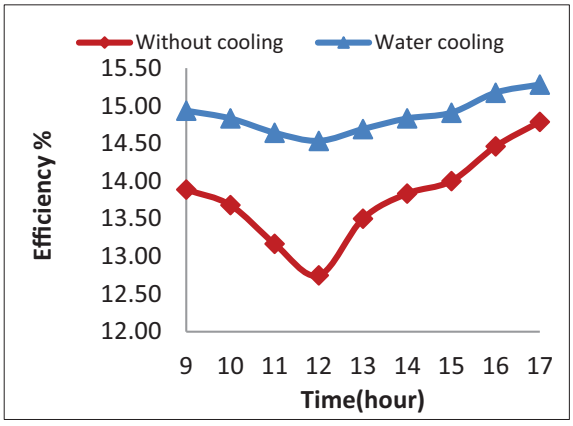


Fig. 12. Comparison of efficiency when using water sprinklers over time.

about 2.9% to 10.8%. Additionally, Figure 11 shows also the best positive influence of hybrid cooling, presenting enhancements in output power between approximately 6% and 13%. These results emphasize the potential of both hybrid cooling systems to positively impact the overall performance of photovoltaic panels in diverse environmental conditions.

2.3.3 Efficiency improvement of PV panels with cooling

The presented data illustrates the efficiency of photovoltaic (PV) panels under two main conditions: without cooling and with cooling. The interpretations of the results will be as follows: Throughout the observed time intervals, there is a consistent trend of higher efficiency with cooling compared to without cooling. This indicates that the cooling system has a positive impact on the overall efficiency of the PV panels. The sustained improvement in efficiency is a significant outcome, suggesting that the cooling system contributes to a more effective conversion of sunlight into electrical energy.

According to Figure 12, the efficiency gains with water cooling, expressed as a percentage, fluctuate across different periods. The values span approximately 0.6% to 1.8%, indicating the extent of enhancement achieved through the application of the water-cooling system. Similarly, Figure 13 illustrates efficiency improvements with air cooling, ranging from about 0.44% to 1.45%, while in Figure 14, the hybrid cooling system demonstrates the highest efficiency increase, expressed as a percentage, varying from approximately 0.88% to 2%, underscoring the positive impact of the hybrid cooling system in enhancing efficiency. These findings emphasize the potential for improved efficiency with the incorporation of different cooling methods.

In summary, the results suggest that the implementation of a cooling system positively influences the efficiency of the PV panels. The observed trends indicate that cooling has a reliable and consistent impact, contributing to improved overall performance and energy conversion efficiency.

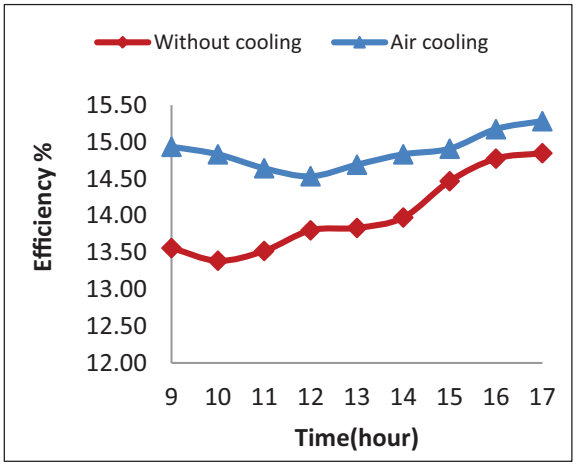


Fig. 13. Comparison of efficiency when using air cooling over time.

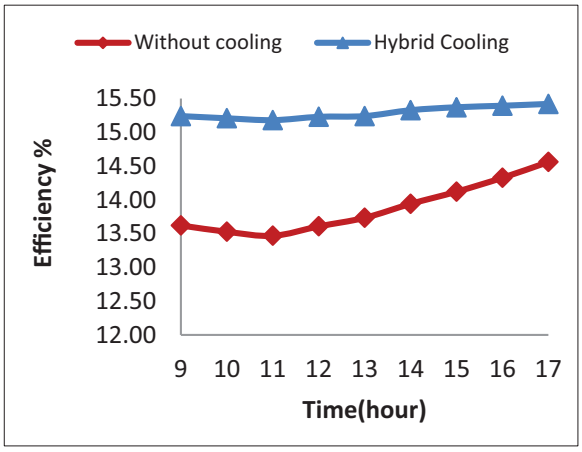


Fig. 14. Comparison of efficiency when using hybrid cooling over time.

3 Comprehensive evaluation of PV cooling technologies

3.1 Cost analysis

The proposed solution, which combines water and air cooling in a hybrid system, may involve initial setup costs for installing the necessary equipment, including water circulation systems, fans or blowers, and temperature sensors. However, due to its effectiveness in enhancing efficiency and longevity while maintaining lower temperatures, the long-term benefits may outweigh the initial investment.

The observed efficiency improvement with hybrid cooling systems stems from the synergistic effects of combining water- and air-cooling methods. By dissipating heat through both conduction and convection, hybrid systems effectively reduce thermal losses and maintain optimal operating temperatures for photovoltaic (PV) panels. Strategic control and optimization of these systems are crucial for maximizing efficiency and energy output. Dynamically adjusting water and air flow rates based on environmental conditions allows stakeholders to enhance cooling performance while minimizing energy consumption. Our study provides valuable insights into cooling system performance, but further research is needed. Long-term durability testing of cooling components and their integration with PV installations is essential to evaluate reliability and maintenance needs. Additionally, optimizing control algorithms and exploring alternative cooling methods represent promising areas for future research. Advanced control strategies using machine learning and artificial intelligence can improve the adaptability of cooling systems. We also assess the impact of ambient temperature and solar radiation on system performance, demonstrating the system's ability to adapt to varying conditions and achieve significant reductions in PV temperature, ultimately boosting power output. A comprehensive summary of related studies highlights the effectiveness of cooling techniques in enhancing power generation (Tab. 4), provides a clearer understanding of the various contributions from researchers in the field, focusing on methods of cooling, temperature reduction, and the associated increase in power.

3.2 Comparison with other PV cooling technologies

Existing PV cooling technologies such as passive cooling methods (e.g., heat sinks, natural convection) and active cooling systems (e.g., liquid cooling, forced air-cooling) also incur installation and maintenance costs. However, their effectiveness in reducing panel temperature and improving efficiency may vary based on factors such as local climate conditions and system design.

3.3 Key results elaboration

In this section, we delve into the key results obtained from our study, discussing their implications, limitations, and avenues for future research.

- Our study has elucidated the effectiveness of various cooling systems in enhancing the performance of photovoltaic (PV) systems. Through experimental validations, we have quantified the temperature reduction and efficiency improvements achieved by water, air, and hybrid cooling methods.
- However, it is essential to acknowledge the limitations of our experimental study. For instance, many overlook real-world factors such as wind effects, which can influence the thermal behavior of PV panels, humidity, and dust.
- Furthermore, the cost implications of each cooling system warrant discussion. While water cooling systems may entail higher initial investment due to infrastructure requirements, they offer superior efficiency and long-term performance compared to air cooling methods. However, the energy consumption of fans utilized in air cooling systems must be considered, as it can impact overall efficiency and operational costs.

Table 4. Comparison of the experimental findings with other literature in standard conditions.

Authors	Coolant	Temperature reduction (°C)	Increase in output power
[38] Ismail et al.,(2023)	Porous media	20.2	9.4
[38] Ismail et al.,(2023)	Porous media	46 (indoor)	50
[39] Ismail et al.,(2021)	Nano-Fluid	50 (indoor)	60
[40] Lin et al., (2019)	Air	4	7
[41] Micheli et al., (2015)	Air	13	7.7
[42] Moradgholi et al., (2014)	Methanol	13	5
[43] Smith et al., (2014)	Water	11	6
[44] Hasan et al., (2016)	PCM	10	6.6
[45] Tang et al., (2010)	Air	4.7	8
[30] Ebaid et al., (2018)	Nano-Fluid	20	9
[46] El Mays et al., (2017)	Air	6	–
[19] Mazón-Hernández et al., (2013)	Air	22	–
[47] Wu et al., (2018)	Air	12.5	–
[48] Ebrahimi et al., (2015)	Air	20	
[49] Amori and Al-Najjar, (2012)	Air	16	
[50] Herez et al., 2020	Heat pipe cooling	14	–
[51] Karami & Rahimi, (2014)	Nano-Fluid	18	
[52] Al-Musawi et al., (2019)	Nano-Fluid	19	
[53] Bahaidarah et al., (2013)	Water	17	–
[54] Rahimi et al., (2014)	Water	26	–
[10] Nižetić et al., (2016)	Water	26	–
[55] Baloch et al., (2015)	Water	12	–
[56] Hachicha et al., (2015)	Water	7.7	–
[57] Irwan et al., (2015)	Water	15	–
[58] Wu & Xioang, (2014)	Water	18.5	–
[59] Barrau et al., (2011)	Water	2	–

4 Conclusions

In conclusion, this study highlights the significant benefits of implementing hybrid cooling systems to enhance the performance and efficiency of photovoltaic (PV) panels. The integration of water- and air-cooling methods proves to be effective, with the hybrid system achieving a notable reduction in panel temperatures by approximately 40%. This temperature decrease directly correlates to an increase in power output of over 13% and a 2% boost in overall efficiency.

The cooling system demonstrated consistent effectiveness throughout the day, particularly during peak solar radiation hours, showcasing its crucial role in mitigating thermal losses. Additionally, our findings indicate that optimizing water and air flow rates can further maximize energy generation, even under high ambient temperatures. Furthermore, adjusting the tilt angle of the solar panels presents an additional opportunity to enhance cooling efficiency and overall energy output.

These results underscore the critical importance of hybrid cooling systems in maximizing the effectiveness and sustainability of solar energy generation. By ensuring sustained efficiency and prolonging the lifespan of PV systems, these cooling solutions represent a promising avenue for improving solar energy applications in hot regions. Overall, this study contributes valuable insights into optimizing solar energy systems, paving the way for future advancements in cooling technologies.

4.1 Limitations and future scope

– Despite the promising results, it is important to acknowledge the limitations of our study, including the simplified experimental setup and focus on specific environmental conditions. Future research should explore the applicability of the proposed cooling system across diverse climatic regions and operational scenarios.
– Furthermore, investigating the long-term performance and durability of the cooling system, as well as its integration with existing PV installations, presents an avenue for future research.

4.2 Practical implications

– The findings of our study have significant practical implications for the solar energy industry, offering a cost-effective solution for improving the performance of PV systems.

– Adopting a hybrid method of solar panels cooling with water and air in hot and arid climates ensures the reduction of excessive heat and the steadiness of the operation of these panels for longer periods, with a notable increase in the overall efficiency of the system.
– By optimizing water, air, and hybrid cooling methods, stakeholders can enhance the reliability and efficiency of solar energy generation, thereby reducing operational costs and carbon emissions.

4.3 Economic benefits

– Implementing hybrid cooling technology presents substantial economic benefits, including increased energy production, reduced maintenance costs, and extended lifespan of PV panels. Moreover, the potential for leveraging tax incentives and subsidies further enhances the economic viability of adopting this technology, making it an attractive investment for homeowners and commercial entities alike.

In summary, our study provides valuable insights into the effectiveness and practical implications of hybrid cooling systems for enhancing the performance of PV systems. We pave the way for the widespread adoption of sustainable and efficient solar energy solutions by addressing key challenges and identifying optimal cooling strategies.

The study reflects ongoing solar panel cooling technology advancements, promising improved efficiency and reliability in renewable energy systems. Effective cooling methods contribute to global efforts in combating climate change and transitioning towards cleaner energy sources. Collaboration across disciplines is essential for addressing complex energy challenges and driving innovation in renewable energy solutions. Adaptable cooling technologies are crucial for maintaining solar panel performance in the face of climate variability and extreme weather events.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Conflicts of interest

The author(s) declared no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author, [A. S. Y. Alsabagh].

Author contribution statement

The authors confirm their contribution to the paper as follows: Experimental work: Ismail Al-Masalha; Data collection: Omar Badran; Writing and analysis: Abdel Salam Alsabagh; reviewing and discussion: Taiseer M. Abu-Rahmeh; Conceptualization: Aiman Al Alawin; Writing and validation: Naim Rizq Alkawaldeh. All authors reviewed the results and approved the final version of the manuscript.

References

1. D. Al-Jamea et al., Investigation on Water Immersing and Spraying for Cooling PV Panel , Int. Rev. Mech. Eng. **16**, 9 (2022)
2. J. Siecker, K. Kusakana, E.B. Numbi, A review of solar photovoltaic systems cooling technologies, Renew. Sustain. Energy Rev. **79**, 192 (2017)
3. A.S. Abdelrazik et al., A review on recent development for the design and packaging of hybrid photovoltaic/thermal (PV/T) solar systems, Renew. Sustain. Energy Rev. **95**, 110 (2018)
4. H.M. Bahaidarah, A.A. Baloch, P. Gandhidasan, Uniform cooling of photovoltaic panels: A review, Renew. Sustain. Energy Rev. **57**, 1520 (2016)
5. M. Azmi et al., Hybrid cooling system for solar photovoltaic panel, J. Phys.: Conf. Ser. **2550**, 012004 (2023)
6. A. Mahdavi et al., A review of passive cooling of photovoltaic devices, Clean. Eng. Technol. **11**, 100579 (2022)
7. S. Panda et al., A review on advanced cooling techniques for photovoltaic panel, Mater. Today: Proc. **62**, 6799 (2022)
8. S.A. Zubeer, H. Mohammed, M. Ilkan. A review of photovoltaic cell cooling techniques, E3S Web Conf. **22**, 00205 (2017)
9. P. Dwivedi et al., Advanced cooling techniques of PV modules: a state of the art, Case Stud. Therm. Eng. **21**, 100674 (2020)
10. S. Nižetić et al., Water spray cooling technique applied on a photovoltaic panel: The performance response, Energy Convers. Manag. **108**, 287 (2016)
11. K.A. Moharram et al., Enhancing the performance of photovoltaic panels by water cooling, Ain Shams Eng. J. **4**, 869 (2013)
12. L.D. Jathar et al., Comprehensive review of environmental factors influencing the performance of photovoltaic panels: concern over emissions at various phases throughout the lifecycle, Environ. Pollut. **326**, 121474 (2023)
13. S.S. Koohestani et al., Comparative review and evaluation of state-of-the-art photovoltaic cooling technologies, J. Cleaner Prod. **406**, 136953 (2023)
14. S. Shalaby et al., Experimental study on the performance of PV with water cooling, Energy Rep. **8**, 957 (2022)
15. H.A. Kazem et al., Enhancement of photovoltaic module performance using passive cooling (Fins): a comprehensive review, Case Stud. Thermal Eng. **49**, 103316 (2023)
16. Z. Arifin, D.D.D.P. Tjahjana, S. Hadi, R.A. Rachmanto, G. Setyohandoko, B. Sutanto, Numerical and experimental investigation of air cooling for photovoltaic panels using aluminum heat sinks, Int. J. Photoenergy (2020)
17. A. Palumbo, Design and Analysis of Cooling Methods for Solar Panels, Master thesis, Mechanical Engineering Department, Youngstown State University (2013)
18. E. Cuce, T. Bali, S.A. Sekucoglu, Effects of passive cooling on performance of silicon photovoltaic cells, Int. J. Low-Carbon Technol. **6**, 299 (2011)

19. R. Mazón-Hernández, J.R. García-Cascales, F. Vera-García, A.S. Káiser, B. Zamora, Improving the electrical parameters of a photovoltaic panel by means of an induced or forced air stream, Int. J. Photoenergy (2013)
20. N. Nader, W. Al-Kouz, S. Al-Dahidi, Assessment of existing photovoltaic system with cooling and cleaning system: case study at Al-Khobar City, Processes **8**, 9 (2019)
21. S.A. Rakino, S. Suherman, S. Hasan, A.H. Rambe, A passive cooling system for increasing efficiency of solar panel output, J. Phys.: Conf. Ser. **1373**, 12017 (2019)
22. C.G. Popovici, S.V. Hudisteanu, T.D. Mateescu, N.-C. Chereches, Efficiency improvement of photovoltaic panels by using air-cooled heat sinks, Energy Proc. **85**, 425 (2016)
23. Z. Farhana, Y.M. Irwan, R.M.N. Azimmi, A.R.N. Razliana, N. Gomesh, Experimental investigation of photovoltaic modules cooling system, in *Proceedings of the 2012 IEEE Symposium on Computers & Informatics (ISCI)* (IEEE), pp. 165–169
24. S. Dubey, G.S. Sandhu, G.N. Tiwari, Analytical expression for electrical efficiency of PV/T hybrid air collector, Appl. Energy **86**, 697 (2009)
25. A. Shahsavar, M. Salmanzadeh, M. Ameri, P. Talebizadeh, Energy saving in buildings by using the exhaust and ventilation air for cooling of photovoltaic panels, Energy Build. **43**, 2219 (2011)
26. P. Dwivedi, S.A. Ganesh, K. Sudhakar, A. Soni, S. Shanmuga Priya, Thermal and electrical performance of uncooled, nature-cooled, and photovoltaic thermal module, Int. J. Photoenergy (2023)
27. A. Durez, M. Ali, A. Waqas, K. Nazir, S. Kumarasamy, Modelling and optimization of phase change materials (PCM)-based passive cooling of solar PV panels in multi climate conditions, Front. Energy Res. **11**, 1121138 (2023)
28. P. Dwivedi, K. Sudhakar, A. Soni, E. Solomin, I. Kirpichnikova, Advanced cooling techniques of PV modules: a state of art, Case Stud. Thermal Eng. **21**, 100674 (2020)
29. A. Hussien, A. Eltayesh, H.M. El-Batsh, Experimental and numerical investigation for PV cooling by forced convection, Alexandria Eng. J. **64**, 427 (2023)
30. M.S. Ebaid et al., Exploring cooling of PV panels based on metallic and nonmetallic nanofluids: an experimental study, Adv. Mech. Eng. **16**, 1 (2024)
31. M. Fan et al., Simulation of air-cooled PV/T air conditioning system for cooling and power cogeneration, Appl. Thermal Eng. **224**, 119971 (2023)
32. Tongwei Co., Ltd. https://en.tongwei.com.cn/
33. M. Aghaei et al., Review of degradation and failure phenomena in photovoltaic modules, Renew. Sustain. Energy Rev. **159**, 112160 (2022)
34. I. Laabab, S. Ziani, A. Benami, A literature review of solar cell overheating control, in *The International Conference on Artificial Intelligence and Smart Environment* (Springer, 2022)
35. C. Olalla et al., Mitigation of hot-spots in photovoltaic systems using distributed power electronics, Energies **11**, 726 (2018)
36. D. Govindasamy, A. Kumar, Evaluation of the impact of different composite phase change materials on reduction in temperature and enhancement of solar panel efficiency, J. Energy Storage **60**, 106631 (2023)
37. E. Arslan et al., Designing of a new type air-water cooled photovoltaic collector, Tehnički Glasnik **14**, 41 (2020). https://doi.org/10.31803/tg-20190227095246
38. I. Masalha et al., Outdoor experimental and numerical simulation of photovoltaic cooling using porous media, Case Stud. Thermal Eng. **42**, 102748 (2023)
39. I. Masalha et al., An experimental and numerical study to improve the efficiency of PV modules by using nano-fluid cooling system, Int. Rev. Mech. Eng. **15**, 582 (2021)
40. Y. Lin et al., 17% efficient organic solar cells based on liquid exfoliated WS2 as a replacement for PEDOT: PSS, Adv. Mater. **31**, 1902965 (2019)
41. A.A. Hachicha, C. Ghenai, A.K. Hamid, Enhancing the performance of a photovoltaic module using different cooling methods, Int. J. Energy Power Eng. **9**, 1106 (2015)
42. M. Moradgholi, S.M. Nowee, I. Abrishamchi, Application of heat pipe in an experimental investigation on a novel photovoltaic/thermal (PV/T) system, Solar Energy **107**, 82 (2014)
43. M.K. Smith et al., Water cooling method to improve the performance of field-mounted, insulated, and concentrating photovoltaic modules, J. Solar Energy Eng. **136**, 034503 (2014)
44. A. Hasan, H. Alnoman, A.H. Shah, Energy efficiency enhancement of photovoltaics by phase change materials through thermal energy recovery, Energies **9**, 782 (2016)
45. X. Tang, Z. Quan, Y. Zhao, Experimental investigation of solar panel cooling by a novel micro heat pipe array, Energy Power Eng. **2**, 171 (2010)
46. A. El Mays et al., Improving photovoltaic panel using a finned plate of aluminum, Energy Proc. **119**, 812 (2017)
47. S.-Y. Wu, C. Chen, L. Xiao, Heat transfer characteristics and performance evaluation of water-cooled PV/T system with cooling channel above PV panel, Renew. Energy **125**, 936 (2018)
48. M. Ebrahimi, M. Rahimi, A. Rahimi, An experimental study on using natural vaporization for cooling of a photovoltaic solar cell, Int. Commun. Heat Mass Transfer **65**, 22 (2015)
49. K.E. Amori, H.M.T. Al-Najjar, Analysis of the thermal and electrical performance of a hybrid (PV/T) air-based solar collector for Iraq, Appl. Energy **98**, 384 (2012)
50. A. Herez et al., Review on photovoltaic/thermal hybrid solar collectors: classifications, applications, and new systems, Solar Energy **207**, 1321 (2020)
51. N. Karami, M. Rahimi, Heat transfer enhancement in a PV cell using Boehmite nanofluid, Energy Convers. Manag. **6**, 275 (2014)
52. A.I.A. Al-Musawi et al., Numerical study of the effects of nanofluids and phase-change materials in photovoltaic thermal (PVT) systems, J. Thermal Anal. Calorim. **137**, 623 (2019)
53. H. Bahaidarah et al., Performance evaluation of a PV (photovoltaic) module by back surface water cooling for hot climatic conditions, Energy **59**, 445 (2013)
54. M. Rahimi et al., Design of a self-adjusted jet impingement system for cooling of photovoltaic cells, Energy Convers. Manag. **83**, 48 (2014)
55. A.A. Baloch et al., Experimental and numerical performance analysis of a converging channel heat exchanger for PV cooling, Energy Convers. Manag. **103**, 14 (2015)

56. A.A. Hachicha, C. Ghenai, A.K. Hamid, Enhancing the performance of a photovoltaic module using different cooling methods. Int. J. Energy Power Eng. **9**, 1106 (2015)
57. Y. Irwan et al., Indoor test performance of PV panel through water cooling method, Energy Proc. **79**, 604 (2015)
58. S. Wu, C. Xiong, Passive cooling technology for photovoltaic panels for domestic houses, Int. J. Low-Carbon Technol. **9**, 118 (2014)
59. J. Barrau et al., Effect of a hybrid jet impingement/micro-channel cooling device on the performance of densely packed PV cells under high concentration, Solar Energy **85**, 2655 (2011)

Cite this article as: Ismail Al-Masalha, Abdel Salam Alsabagh, Omar Badran, Naim Alkawaldeh, Taiseer M. Abu-Rahmeh, Aiman Al Alawin, Improving photovoltaic module efficiency using water sprinklers, air fans, and combined cooling systems, EPJ Photovoltaics **15**, 41 (2024)